

Samia Carchaf . Jarvis Consulting

I recently completed a Master's degree in Computer Science with a specialization in Artificial Intelligence from UQAC and a Software Engineering degree from Polytech Tours, and have been building hands-on experience in machine learning and data engineering through projects at Jarvis Consulting Group. My background includes working with Python, SQL, PostgreSQL, Docker, Bash scripting, and Linux to build data collection, automation, and infrastructure solutions. What excites me most about the AI and machine learning industry is the opportunity to solve real business problems with technology while continuously learning new tools and systems. I am especially passionate about the intersection of machine learning and data engineering, combining analytical thinking with the development of practical and scalable solutions.

Skills

Proficient: Python, Linux/Bash, RDBMS/SQL, Docker, Git

Competent: Machine Learning, Agile/Scrum, Power BI, Data Modeling, Bash Scripting

Familiar: Java, Azure ML, AWS, MLflow, Tableau

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_SamiaCarchaf

Cluster Monitor [GitHub]:

- Designed and implemented a Linux Cluster Monitoring Agent to collect hardware specifications and real-time resource usage data from Linux servers.
- Provisioned a PostgreSQL database with Docker for persistent storage.
- Automated continuous data collection every minute using crontab, enabling real-time cluster monitoring for the LCA team.

RDBMS and SQL [GitHub]:

- Designed and implemented a relational database schema using PostgreSQL, applying normalization principles and enforcing data integrity through primary keys, foreign keys, and constraints.
- Wrote optimized SQL queries to answer business questions.

Python Data Analytics [GitHub]:

- Built an end-to-end Python analytics pipeline on the U.S. CFPB Consumer Complaint Database.
- Conducted data profiling, cleaning, and feature engineering with Pandas, including company-tier segmentation and time-series trends.
- Produced interactive visualizations across 7 key findings, including market concentration, product mix evolution, and a seven-dimension Bureau Accountability Scorecard.

Highlighted Projects

Credit Card Fraud Detection [GitHub]: Developed a fraud detection system analyzing 590K+ transactions from the IEEE-CIS dataset using XGBoost. Addressed extreme class imbalance (0.3% fraud rate) using SMOTE and class weight adjustment. Designed a Power BI dashboard for real-time monitoring of suspicious transactions. Technologies used: Python, XGBoost, SMOTE, scikit-learn, Pandas, Power BI.

Professional Experiences

Machine Learning Engineer, Jarvis (2026-present): Collaborated within Agile/Scrum teams on diverse software and data-driven projects.

Data Infrastructure Manager, Médecins francophones du Canada (2026-01 to 2026-04): Designed and deployed the organization's core data architecture, migrating historical data into a structured SQL environment. Built automated ETL pipelines between CiviCRM and Power BI, reducing report production time by 80%. Created interactive Power BI dashboards for predictive analysis of membership trends and KPI monitoring.

Data Scientist Intern, Francoflex (2025-07 to 2025-12): Developed an end-to-end ML pipeline for accented French speech recognition, processing 3K+ audio samples. Improved model accuracy by 35% through data augmentation techniques. Set up a replicable training infrastructure using Python, Docker, and MLflow.

Data Analyst Intern, ARSENE (2024-06 to 2024-08): Optimized complex SQL queries reducing execution times by 25%. Conducted data quality analysis identifying 20+ critical anomalies in ETL pipelines. Built internal dashboards to monitor data quality using Python and SQL.

Education

University of Québec at Chicoutimi (UQAC) (2024-2026), Master of Science, Computer Science - Artificial Intelligence - Specializations: Machine Learning, NLP, Deep Learning, Data Mining

Polytech Tours Engineering School (2020-2025), Engineering Degree, Software Engineering - Focus: Big Data, Statistics, Algorithms, Relational Databases

Miscellaneous

- Power BI (DataCamp): Data visualization and DAX proficiency
- Fraud Detection in Python (DataCamp): ML application using XGBoost and SMOTE
- Data Analytics (IBM): ID: 25UDNA8FVK48
- French (Native)
- English (Professional - TOEIC 890/990)